

STAAR-Aligned Math Lesson Plans — Grades 3-5

5 Fun & Engaging Lessons with STAAR 2.0 Practice Built In

Lesson 1: Math Mystery Detective

Grade: 3rd **TEKS:** 3.4K (Solve one-step and two-step problems involving multiplication and division), 3.5B (Represent and solve one- and two-step multiplication and division problems within 100) **Duration:** 60 minutes **Materials:** Mystery case folders (printed), magnifying glasses (optional/fun), dry erase boards, markers, "Evidence Logs" (printed), manipulatives (counters, base-ten blocks)

Learning Objective

I can solve multiplication and division word problems by identifying clues in the text and choosing the correct operation.

STAAR Connection

This lesson practices **multi-step problem solving** and **constructed response** — both key STAAR 2.0 formats. Students must explain their thinking, not just find an answer.

Warm-Up (10 minutes): "Detective Training"

Teacher says: "Today you're becoming Math Mystery Detectives! But first, every detective needs training."

Display these "Quick Cases" one at a time:

1. "There are 4 teams. Each team has 6 players. How many players in all?" ($4 \times 6 = 24$)
2. "24 stickers shared equally among 3 friends. How many each?" ($24 \div 3 = 8$)
3. "A baker made 5 trays of cookies. Each tray has 7 cookies. He gave away 10 cookies. How many are left?" ($5 \times 7 = 35$; $35 - 10 = 25$)

Students solve on dry erase boards, hold up on signal. **Key:** For #3, ask "What makes this problem HARDER than #1 and #2?" (two steps!)

Main Activity (25 minutes): "The Case of the Missing Math"

Setup: Divide class into teams of 3-4 students. Each team receives a "Mystery Case Folder" containing 4 mystery word problems. Each problem is a "case" they must "solve."

The Mystery Cases:

Case #1: The Pet Store Puzzle

Mr. Garcia's pet store received 8 boxes of dog treats. Each box has 9 bags inside. He put the bags on 6 shelves equally. How many bags are on each shelf?

Show your detective work:

- What operation(s) do you need? _____
- Step 1: _____
- Step 2: _____
- Answer with a complete sentence: _____

Solution: $8 \times 9 = 72$ bags total; $72 \div 6 = 12$ bags per shelf

Case #2: The Lunch Room Mystery

The cafeteria ordered 56 cartons of milk for 7 classes. Each class got the same number. Then 3 cartons from Mrs. Lee's class were returned because they were expired. How many good cartons does Mrs. Lee's class have?

Detective Evidence Log:

- Draw a picture or model to represent the problem
- Write an equation for each step
- Explain in words why you chose those operations

Solution: $56 \div 7 = 8$ cartons per class; $8 - 3 = 5$ good cartons

Case #3: The Field Trip Fiasco

48 students are going on a field trip. Each van holds 8 students. The school already has 2 vans. How many MORE vans does the school need to rent?

Constructed Response (STAAR format): How many vans are needed in all? Show your work and explain your reasoning. Then determine how many vans must be rented.

Solution: $48 \div 8 = 6$ vans needed; $6 - 2 = 4$ more vans to rent

Case #4: The Birthday Party Problem (*Challenge Case*)

Mia is buying party favors for her birthday. She invited 9 friends. She wants to give each friend 3 stickers, 2 erasers, and 1 pencil. Stickers come in packs of 9. Erasers come in packs of 6. How many packs of stickers and how many packs of erasers does she need?

Multi-Select (STAAR format): Select ALL the equations you would need to solve this problem:

- ☐ $9 \times 3 = 27$
- ☐ $9 + 3 = 12$
- ☐ $27 \div 9 = 3$
- ☐ $9 \times 2 = 18$
- ☐ $18 \div 6 = 3$
- ☐ $18 - 6 = 12$

Correct: $9 \times 3 = 27$; $27 \div 9 = 3$ packs of stickers; $9 \times 2 = 18$; $18 \div 6 = 3$ packs of erasers

Process: Teams work through cases together. Teacher circulates, asking guiding questions:

- "What's happening in this problem?"
- "What clue words help you decide the operation?"
- "Is this a one-step or two-step problem? How do you know?"

Debrief: Each team presents one case to the class, explaining their "detective work." Class discusses different solution strategies.

Practice (15 minutes): "Solo Detective Work"

Students independently solve 2 problems in STAAR format:

Problem 1 (Constructed Response):

Marcus has 7 bags of marbles. Each bag has 8 marbles. He wants to share all the marbles equally among himself and 3 friends. How many marbles will each person get?

Show your work. Use numbers, words, or pictures to explain your answer.

Problem 2 (Multi-Select):

A teacher bought 6 packs of pencils with 10 pencils in each pack. She gave 4 pencils to each of her 12 students. Which statements about the leftover pencils are true? Select TWO.

- ☐ There are 12 leftover pencils
- ☐ There are 60 pencils in all
- ☐ The students received 48 pencils total
- ☐ There are no leftover pencils
- ☐ There are 8 leftover pencils

Correct: 60 pencils total (6×10); 48 given away (4×12); $60 - 48 = 12$ leftover

Exit Ticket (5 minutes)

There are 45 apples to put into bags. Each bag holds 5 apples. The store needs 3 more bags than that for oranges. How many bags does the store need altogether?

Write your equation(s) and answer in a complete sentence.

Solution: $45 \div 5 = 9$ apple bags; $9 + 3 = 12$ orange bags; $9 + 12 = 21$ bags total

Differentiation

Group

Modification

ELL Provide a "Math Detective Vocabulary Card" with clue words: *each, every, equally, shared, total, in all, how many more* with visuals. Allow students to draw models. Pair with a bilingual buddy.

SpEd Reduce to 2 cases instead of 4. Provide sentence stems: "First, I need to ____." Then I need to ____." Allow use of manipulatives (counters) throughout. Provide a step-by-step problem-solving checklist.

GT Add a "Master Case" requiring 3 steps. Challenge: "Create your own Mystery Case for a classmate to solve." Must include at least 2 operations.

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Lesson 2: Fraction Pizza Party

Grade: 4th **TEKS:** 4.3A (Represent a fraction a/b as a sum of fractions $1/b$), 4.3B (Decompose a fraction in more than one way), 4.3C (Determine if two given fractions are equivalent using comparisons) **Duration:** 60 minutes **Materials:** Paper plate "pizzas" (1 per student), markers/crayons, fraction strips, "Pizza Order" cards (printed), scissors, STAAR practice sheet

Learning Objective

I can decompose fractions, add fractions with like denominators, and identify equivalent fractions using pizza models.

STAAR Connection

STAAR 2.0 requires students to **model fractions visually** and **explain reasoning** about fraction relationships — not just compute.

Warm-Up (8 minutes): "Pizza Survey"

Teacher says: *"We're opening a class pizzeria today! But first, I need to know your preferences."*

Display: "If you could have $\frac{6}{8}$ of a pizza, how many different ways could you split that into slices?"

Students work with a partner to find at least 2 ways:

- $\frac{6}{8} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$
- $\frac{6}{8} = \frac{2}{8} + \frac{4}{8}$
- $\frac{6}{8} = \frac{3}{8} + \frac{3}{8}$

Share out. Record student thinking on the board. **Ask:** "Did anyone notice that $\frac{6}{8}$ is the same as another fraction we know?" ($\frac{3}{4}$)

Main Activity (25 minutes): "Pizza Party Challenge"

Phase 1: Build Your Pizza (10 min)

Each student gets a paper plate "pizza." They divide it into 8 equal slices by folding/drawing.

Teacher guides:

1. "Color $\frac{3}{8}$ of your pizza with pepperoni (red)."
2. "Color $\frac{2}{8}$ of your pizza with mushrooms (brown)."
3. "How much of your pizza has toppings?" ($\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$)
4. "How much is plain?" ($\frac{3}{8}$)
5. "Write an equation that shows all the parts of your pizza equal one whole." →
 $\frac{3}{8} + \frac{2}{8} + \frac{3}{8} = \frac{8}{8} = 1$

Phase 2: Pizza Orders (15 min)

Teams of 3-4 receive "Pizza Order Cards." Each card describes a customer order that requires fraction thinking:

Order #1: The Picky Eater

"I want exactly $\frac{1}{2}$ of my pizza with cheese only. But my pizza is cut into 8 slices. How many slices should be cheese only? Draw it on your pizza."

Is $\frac{4}{8}$ equivalent to $\frac{1}{2}$? How do you know? Prove it TWO different ways.

Order #2: The Sharing Family

"Our family of 4 ordered a pizza cut into 12 slices. Each person should get the same amount. What fraction of the pizza does each person get? Simplify your answer."

Each person gets $\frac{3}{12} = \frac{1}{4}$. **STAAR Constructed Response:** "Explain why $\frac{3}{12}$ and $\frac{1}{4}$ represent the same amount of pizza. Use words, numbers, or pictures."

Order #3: The Party Planner

"I need $\frac{5}{6}$ of a pizza for my guests. Show three different ways I could decompose $\frac{5}{6}$."

Examples: $\frac{1}{6} + \frac{4}{6}$; $\frac{2}{6} + \frac{3}{6}$; $\frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{2}{6}$

Order #4: The Comparison Challenge

"Two pizzas are the same size. Pizza A has $\frac{3}{4}$ eaten. Pizza B has $\frac{5}{8}$ eaten. Which pizza has more left over?"

Multi-Select: Select ALL true statements:

- ☐ Pizza A has $\frac{1}{4}$ left
- ☐ Pizza B has $\frac{3}{8}$ left
- ☐ Pizza B has more left than Pizza A
- ☐ $\frac{1}{4}$ is the same as $\frac{2}{8}$
- ☐ Pizza A has more left than Pizza B

Correct: Pizza A has $\frac{1}{4}$ left ✓; Pizza B has $\frac{3}{8}$ left ✓; Pizza B has more left ✓; $\frac{1}{4} = \frac{2}{8}$ ✓

Practice (17 minutes): "Menu Math"

Students work independently on a "Menu Math" sheet:

1. **Decompose $\frac{7}{10}$ three different ways.** Show each way with a picture and an equation.
2. **Constructed Response:** Jayla ate $\frac{2}{6}$ of a pie. Her brother ate $\frac{3}{6}$. Jayla says they ate the same amount because 2 and 3 are close together. Is Jayla correct? Explain why or why not using pictures, numbers, or words.

3. **Multi-Select:** Which of the following fractions are equivalent to $\frac{1}{2}$? Select ALL that apply.

- ☐ $\frac{2}{4}$
- ☐ $\frac{3}{5}$
- ☐ $\frac{4}{8}$
- ☐ $\frac{5}{10}$
- ☐ $\frac{3}{6}$
- ☐ $\frac{2}{6}$

Correct: $\frac{2}{4}$, $\frac{4}{8}$, $\frac{5}{10}$, $\frac{3}{6}$

4. **Draw and solve:** A pizza is cut into 12 equal slices. You eat 3 slices. Your friend eats 5 slices. What fraction of the pizza is left? Write an equation to match your picture. $\rightarrow \frac{3}{12} + \frac{5}{12} = \frac{8}{12}$ eaten; $\frac{12}{12} - \frac{8}{12} = \frac{4}{12} = \frac{1}{3}$ left

Exit Ticket (5 minutes)

Part A: Decompose $\frac{4}{6}$ two different ways.

Part B: Is $\frac{4}{6}$ equivalent to $\frac{2}{3}$? Draw a picture to prove your answer.

Differentiation

Group

Modification

- ELL** Provide fraction vocabulary card (numerator, denominator, equivalent, decompose) in English and Spanish. Use real pizza visuals. Pre-cut fraction circles available.
- SpEd** Use physical fraction manipulatives instead of drawing. Provide a completed example for each task type. Reduce Order Cards to 2. Use denominators of 2, 4, and 8 only (easier to visualize).
- GT** Add improper fractions: "If you ate $\frac{11}{8}$ of pizza, what does that mean? Draw it." Challenge: "Find ALL the ways to decompose $\frac{5}{8}$ as a sum of unit fractions and non-unit fractions."
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Lesson 3: Measurement Olympics

Grade: 5th **TEKS:** 5.7A (Solve problems by calculating conversions within a measurement system — customary or metric) **Duration:** 60 minutes **Materials:** Rulers, yardsticks, measuring tape, scale/balance, graduated cylinders, water, small objects to measure, conversion reference cards, "Olympic Record Sheets" (printed), dry erase boards

Learning Objective

I can convert between measurement units within the same system (customary and metric) to solve real-world problems.

STAAR Connection

STAAR frequently asks students to **convert units to compare or solve problems** — not just memorize conversion facts. This lesson emphasizes application.

Warm-Up (8 minutes): "Would You Rather?"

Display these "Would You Rather" questions. Students discuss with a partner, then vote:

1. "Would you rather run 5,000 meters or 3 miles?" ($5,000\text{m} = \sim 3.1\text{ mi}$ — nearly the same!)
2. "Would you rather drink 2 liters of water or 8 cups?" ($2\text{L} \approx 8.45\text{ cups}$ — nearly the same!)
3. "Would you rather carry 10 pounds or 5,000 grams?" ($5,000\text{g} = 5\text{kg} \approx 11\text{ lbs}$ — the grams are slightly heavier!)

Key discussion: "How did you figure out which was more? You had to CONVERT to compare! That's what today is all about."

Post the conversion reference:

Customary:

- 1 foot = 12 inches
- 1 yard = 3 feet
- 1 mile = 5,280 feet
- 1 pound = 16 ounces
- 1 gallon = 4 quarts
- 1 quart = 2 pints
- 1 pint = 2 cups

Metric:

- 1 kilometer = 1,000 meters
 - 1 meter = 100 centimeters
 - 1 kilogram = 1,000 grams
 - 1 liter = 1,000 milliliters
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Main Activity (25 minutes): "Measurement Olympics"

Setup: Set up 5 stations around the room. Teams of 4-5 rotate through stations, spending 5 minutes at each.

Station 1: The Long Jump (Length)

Measure the "long jump" distance marked on the floor (pre-marked with tape at 7 feet, 6 inches).

- Record in inches: _____ inches
- Convert to feet and inches: _____ ft _____ in
- Convert the total inches to centimeters (1 in = 2.54 cm): _____ cm
- **STAAR Challenge:** An Olympic long jumper jumped 8.9 meters. How many centimeters is that? Is the classroom long jump closer to 2 meters or 3 meters?

Solution: 7 ft 6 in = 90 inches; $90 \times 2.54 = 228.6 \text{ cm} = 2.286 \text{ m} \rightarrow \text{closer to 2 meters}$

Station 2: The Weight Lift (Mass/Weight)

Weigh three classroom objects on the scale.

- Object A: _____ grams $\rightarrow$ Convert to kilograms: _____ kg
- Object B: _____ grams $\rightarrow$ Convert to kilograms: _____ kg
- Object C: _____ grams $\rightarrow$ Convert to kilograms: _____ kg
- Total weight: _____ grams = _____ kilograms
- **STAAR Challenge:** A bag of flour weighs 2.5 kilograms. How many grams is that? If a recipe calls for 750 grams, how many recipes can you make with one bag?

Solution: 2,500 grams; $2,500 \div 750 = 3.33 \rightarrow 3 \text{ complete recipes}$

Station 3: The Hydration Station (Capacity)

Using graduated cylinders and water:

- Pour exactly 500 mL of water. That's _____ liters.
- Pour exactly 1.5 liters. That's _____ mL.
- **STAAR Challenge:** An athlete drinks 3 liters of water during a race. Water bottles hold 500 mL each. How many bottles is that?

Solution: $3,000 \text{ mL} \div 500 \text{ mL} = 6 \text{ bottles}$

Station 4: The Sprint Calculator (Multi-Step)

No measuring — pure problem solving!

1. A sprinter runs 400 meters. How many kilometers is 400 meters? If she runs the track 10 times, how many kilometers total?
2. A marathon is 26 miles, 385 yards. How many total yards is that? (1 mile = 1,760 yards)

3. **Constructed Response:** Explain in words how you convert a larger unit to a smaller unit. Give an example.

Solutions: 0.4 km; 4 km; $26 \times 1,760 + 385 = 46,145$ yards

Station 5: The Relay Race (Timed Challenge)

Speed round! Convert as many as you can in 5 minutes:

- 3 yards = _____ feet (9)
- 48 inches = _____ feet (4)
- 5,000 grams = _____ kg (5)
- 2.5 liters = _____ mL (2,500)
- 3 miles = _____ feet (15,840)
- 7 pints = _____ cups (14)
- 4,500 meters = _____ km (4.5)
- 32 ounces = _____ pounds (2)
- 150 cm = _____ meters (1.5)
- 3 gallons = _____ quarts (12)

Practice (17 minutes): STAAR-Style Problems

Problem 1 (Constructed Response):

Coach Williams set up a relay race that is 2 kilometers long. There are 8 runners on the team, and each runner runs the same distance. How many meters does each runner run?

Show your work. Explain how you converted kilometers to meters.

Solution: $2 \text{ km} = 2,000 \text{ m}$; $2,000 \div 8 = 250 \text{ meters each}$

Problem 2 (Multi-Select):

Which of the following are equivalent to 5 feet? Select ALL that apply.

- ☐ 60 inches
- ☐ 1 yard 2 feet
- ☐ 50 inches
- ☐ 1 yard 24 inches
- ☐ 0.5 yards

Correct: 60 inches ✓; 1 yard 2 feet ($3+2=5$) ✓; 1 yard 24 inches ($36+24=60 \text{ in}=5 \text{ ft}$) ✓

Problem 3 (Gridded Response):

Maria has 3 gallons of lemonade for a party. She pours it into cups that each hold 1 pint. How many cups can she fill?

Enter your answer: _____

Solution: 3 gal = 12 qt = 24 pints = 24 cups

Exit Ticket (5 minutes)

A recipe calls for 2 quarts of broth. You only have a measuring cup that measures in cups.

Part A: How many cups of broth do you need? Show your conversion.

Part B: If you double the recipe, how many cups do you need?

Solution: 2 qt = 4 pints = 8 cups; doubled = 16 cups

Differentiation

Group	Modification
ELL	Provide bilingual conversion chart. Use hands-on measuring at every station (concrete before abstract). Pair with English-proficient partner. Vocabulary support: "convert," "equivalent," "unit."
SpEd	Provide completed conversion chart (no memorization required). Reduce stations to 3. Allow calculator for multi-step problems. Provide step-by-step conversion process card: "Step 1: What unit do I have? Step 2: What unit do I need? Step 3: Multiply or divide?"
GT	Add cross-system estimation challenges (e.g., "About how many centimeters in a foot?"). Create their own "Olympic Event" with a conversion problem for classmates. Include unit rate problems.

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Lesson 4: Data Detectives

Grade: 3rd-5th (multi-grade adaptable) **TEKS:** 3.8A (Summarize a data set with multiple categories using a frequency table, dot plot, pictograph, or bar graph with scaled intervals), 4.9A (Represent data on a frequency table, dot plot, or stem-and-leaf plot marked with whole numbers and fractions), 5.9A (Represent categorical and numerical data with dot plots or stem-and-leaf plots) **Duration:** 60 minutes

Materials: "Data Detective Case Files" (printed), graph paper, colored pencils, rulers, class survey materials, sticky notes, dice (optional)

Learning Objective

I can collect, organize, and represent real-world data using appropriate graphs, and draw conclusions from the data.

STAAR Connection

STAAR 2.0 asks students to **interpret data displays and draw conclusions** — including reading graphs they didn't create and answering questions about what the data means.

Warm-Up (8 minutes): "Quick Survey"

Teacher says: *"Data is everywhere! Let's collect some right now."*

Quick class survey: "How many hours of screen time did you have yesterday?"
(Round to nearest hour)

Students write their answer on a sticky note and place it on a class number line (0-10) on the board.

Discuss:

- "What do you notice about our data?"
 - "What's the most common answer?" (mode)
 - "What's the range?"
 - "Are there any outliers?"
 - "If someone asked you 'How much screen time does our class usually have?' what would you say? How do you know?"
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Main Activity (25 minutes): "Data Detective Case Files"

Students work in pairs. Each pair receives a "Case File" with real-world data scenarios.

Case File #1: The School Lunch Investigation

The cafeteria tracked lunch choices for one week:

Day	Pizza	Chicken	Salad	Sandwich
Monday	45	30	15	20
Tuesday	50	25	18	22
Wednesday	40	35	20	15
Thursday	55	28	12	25
Friday	60	20	10	30

Your Mission:

1. Create a bar graph showing the TOTAL of each lunch choice for the week
2. Which lunch was most popular overall? By how many?
3. **Constructed Response:** The cafeteria wants to reduce food waste. Based on this data, what recommendation would you make about how much of each food to prepare? Use specific numbers from the data to support your answer.

Case File #2: The Weather Watchers

Here are the high temperatures (°F) for two weeks: Week 1: 72, 75, 68, 70, 74, 78, 80 Week 2: 65, 62, 68, 70, 63, 60, 66

Your Mission:

1. Create a dot plot for EACH week on the same number line (use different colors)
2. Find the range for each week
3. What's the median temperature for each week?
4. **Multi-Select:** Which statements about the data are true? Select ALL.
 - ☐ Week 1 was generally warmer than Week 2
 - ☐ The range of Week 1 is greater than Week 2
 - ☐ The median of Week 1 is higher than Week 2
 - ☐ 68°F appeared in both weeks
 - ☐ Week 2 had more consistent temperatures

Correct: Week 1 warmer ✓; Week 1 range is 12 vs Week 2 range is 10, so Week 1 range IS greater ✓; Week 1 median (74) > Week 2 median (65) ✓; 68 in both ✓

Case File #3: The Allowance Survey (Grade 4-5)

Students surveyed their classmates about weekly allowance: \$0, \$2, \$5, \$5, \$5, \$3, \$10, \$5, \$7, \$3, \$0, \$5, \$8, \$2, \$5, \$20, \$3, \$5, \$7, \$0

Your Mission:

1. Create a frequency table
2. Create a dot plot
3. Identify the mode, median, and range
4. **Constructed Response:** One student says "most kids get \$5 a week." Another says "the average is about \$5." Are both students correct? Use the data to explain your reasoning.

Mode = \$5 (appears 7 times); Mean = $\$100 \div 20 = \5 ; Median = \$5; Range = \$20; Both are correct!

Practice (17 minutes): STAAR-Format Problems

Problem 1: (Interpret a given graph)

The bar graph shows the number of books students read during the reading challenge:

0-2 books:  (8 students) 3-5 books:  (16 students) 6-8 books:  (12 students) 9+ books:  (4 students)

a) How many students participated in the reading challenge? (40) b) What fraction of students read 6 or more books? ($16/40 = 2/5$) c) **Constructed**

Response: The school wants to set a goal for next year. What would be a reasonable goal for the number of books each student should read? Use the data to justify your answer.

Problem 2 (Multi-Select):

A dot plot shows the number of pets owned by students in class: 0:

●●●●● (5) 1: ●●●●●●● (7) 2: ●●●●●●●●● (9) 3: ●●●● (4) 4: ●● (2) 5: ● (1)

Which conclusions are supported by the data? Select ALL.

- ☐ Most students have 2 pets
- ☐ The range is 5
- ☐ More than half the class has 2 or fewer pets
- ☐ The median number of pets is 2
- ☐ Every student has at least one pet

Correct: Most have 2 ✓; Range is 5 ✓; 21/28 have ≤ 2 pets ✓; Median with 28 data points $\rightarrow$ 14th value = 2 ✓; Not every student has one (5 have 0) ✗

Exit Ticket (5 minutes)

Here are test scores for a group of students: 85, 90, 75, 90, 80, 95, 90, 70, 85, 90

Part A: What is the mode? ____ Part B: Would the mean or the mode better describe the "typical" score? Explain why.

Differentiation**Group****Modification**

ELL Provide data vocabulary cards with visuals (bar graph, dot plot, frequency, mode, median, range). Use picture-based data (food, animals) not abstract numbers. Sentence stems for conclusions: "The data shows that ____."

SpEd Provide pre-made graph templates (axes labeled, scales marked). Reduce data sets to 10 data points. Allow use of calculator for finding mean. Use physical

GT

manipulatives (counters) to build dot plots before drawing.

Include stem-and-leaf plots and histograms. Ask them to find and explain the impact of outliers on mean vs. median. Challenge: "Collect your own data set, create 2 different displays, and explain which display best represents the data and why."

Grade-Level Adjustments

- **3rd Grade:** Focus on Case Files #1 and #2. Use pictographs and bar graphs only. Skip median calculations.
- **4th Grade:** All Case Files. Include dot plots and frequency tables. Introduce median.
- **5th Grade:** All Case Files plus stem-and-leaf plots. Include all measures of center. Add the outlier analysis from GT extensions.

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Lesson 5: Geometry Scavenger Hunt

Grade: 4th **TEKS:** 4.6A (Identify points, lines, line segments, rays, angles, and perpendicular and parallel lines), 4.6B (Identify and draw one or more lines of symmetry, if they exist, for a two-dimensional figure), 4.6C (Apply knowledge of right angles to identify acute, right, and obtuse triangles) **Duration:** 60 minutes

Materials: Protractors (1 per student), rulers, "Scavenger Hunt Checklist" (printed), clipboards, cameras/tablets (optional — for photographing findings), pattern blocks, "Geometry Detective Badge" stickers (optional/fun), colored pencils

Learning Objective

I can identify geometric elements (angles, lines, symmetry) in real-world objects and classify triangles by their angles.

STAAR Connection

STAAR tests geometric vocabulary and classification — students must **identify and describe** geometric features, not just memorize definitions.

Warm-Up (8 minutes): "Geometry Vocabulary Lightning Round"

Display images one at a time. Students hold up the correct vocabulary card (pre-distributed: point, line, line segment, ray, parallel, perpendicular, acute, right, obtuse, symmetry).

1. Railroad tracks going into the distance → **parallel lines**
2. The corner of a book → **right angle**
3. A flashlight beam → **ray**
4. A butterfly → **line of symmetry**
5. The letter X → **intersecting lines**
6. A pizza slice tip → **acute angle**
7. An open book lying flat → **obtuse angle**
8. A plus sign (+) → **perpendicular lines**

Quick discussion: "What's the difference between a line, a line segment, and a ray?"
 (Line goes forever both ways; segment has two endpoints; ray has one endpoint and goes forever one way)

Main Activity (25 minutes): "Geometry Scavenger Hunt"

Phase 1: Classroom Hunt (15 min)

Teams of 3-4 receive a "Scavenger Hunt Checklist" on clipboards. They search the classroom (and hallway, if permitted) for real-world examples of each geometric element.

Scavenger Hunt Checklist:

Find and sketch/describe an example of each:

Geometry Element Where I Found It Sketch ✓

Parallel lines	☐
Perpendicular lines	☐
Acute angle	☐
Right angle	☐
Obtuse angle	☐
Line of symmetry	☐
Ray	☐
Line segment	☐
Acute triangle	☐
Right triangle	☐
Obtuse triangle	☐

BONUS challenges (5 points each):

- ☐ Find an object with EXACTLY 2 lines of symmetry
- ☐ Find an object with MORE THAN 4 lines of symmetry
- ☐ Find a shape with NO lines of symmetry
- ☐ Find two different types of angles on the SAME object
- ☐ Find parallel AND perpendicular lines on the SAME object

Scoring: 1 point per item found; 5 points per bonus; team with most points earns "Geometry Detective" badges.

Phase 2: Measure & Classify (10 min)

Back at desks, students use protractors to measure angles they found:

Activity: Draw 5 angles from the scavenger hunt. Measure each with a protractor. Classify as acute ($<90^\circ$), right ($=90^\circ$), or obtuse ($>90^\circ$).

Angle Source Object Measurement Classification

1	°
2	°
3	°
4	°
5	°

Triangle Classification Challenge: Draw one of each triangle type. Measure all angles in each triangle. What do the angles add up to?

- Acute triangle: All angles $< 90^\circ$; sum = _____°
- Right triangle: One angle = 90° ; sum = _____°
- Obtuse triangle: One angle $> 90^\circ$; sum = _____°

Discovery question: "What do you notice about the sum of angles in EVERY triangle?" (Always 180° !)

Practice (17 minutes): STAAR-Format Problems

Problem 1 (Multi-Select):

Look at the shape below (rectangle drawn). Select ALL statements that are true about this shape.

- ☐ It has 4 right angles
- ☐ It has 2 pairs of parallel sides
- ☐ It has exactly 1 line of symmetry
- ☐ It has perpendicular sides
- ☐ It has 2 lines of symmetry
- ☐ It has 4 acute angles

Correct: 4 right angles ✓; 2 pairs parallel ✓; perpendicular sides ✓; 2 lines of symmetry ✓

Problem 2 (Constructed Response):

Maya says that a triangle with angles measuring 60° , 70° , and 50° is an acute triangle. David says it's an obtuse triangle because 70° is a big

number.

Who is correct? Explain your reasoning. What makes a triangle acute, right, or obtuse?

Problem 3:

Draw ALL lines of symmetry for each shape: a) Equilateral triangle (3 lines) b) Rectangle (2 lines) c) Regular pentagon (5 lines) d) Scalene triangle (0 lines — trick!)

Problem 4 (Multi-Select):

Which of the following shapes have at least one pair of perpendicular sides? Select ALL.

- ☐ Square
- ☐ Rhombus (non-square)
- ☐ Rectangle
- ☐ Right triangle
- ☐ Regular hexagon

Correct: Square ✓; Rectangle ✓; Right triangle ✓

Exit Ticket (5 minutes)

Part A: A triangle has angles of 35° , 55° , and 90° . What type of triangle is this? How do you know?

Part B: Draw a shape that has exactly 2 lines of symmetry AND contains at least one pair of parallel lines. Name your shape.

Part A: Right triangle (has a 90° angle); Part B: Rectangle

Differentiation

Group

Modification

ELL

Provide illustrated geometry vocabulary card in English/Spanish. Use color coding: red = acute, blue = right, green = obtuse. Pair with English-proficient partner for scavenger hunt. Allow labeling in native language.

SpEd

Provide a reference sheet with angle types and measurements. Pre-identify 5 locations in the classroom for them to check (reduce searching overwhelm). Use digital protractor app if physical protractor is difficult. Reduce scavenger hunt to 6 items.

GT

Include "Can you find an angle that measures exactly $___\circ$?" challenges. Research and explain: why do the angles of a triangle always add up to 180° ? Extend to 3D shapes: "Where do you see parallel and perpendicular lines in

3D objects?" Challenge: classify quadrilaterals by their angle and line properties.

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